

**Cross-sectional regression equity multi-factor models:
implementation pitfalls.**

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Cross-sectional regression models

A factor model expressing the cross-section of total returns in terms of factors can be mathematically formulated as (see e.g. R. Haugen, N. Baker. “Commonality in the determinants of expected stock returns.” *Journal of Financial Economics*, Vol. 41, No. 3 (February 1996), pp. 401-439):

$$\mathbf{R}(t) = \mathbf{B}(t-1)\mathbf{F}(t) + \mathbf{u}(t) \quad (1)$$

Where $\mathbf{R}(t)$ is the vector of stock returns from time $t-1$ to time t , $\mathbf{B}(t-1)$ is a $N \times K$ matrix of the exposures of N stocks to K factors that are available as of $t-1$, $\mathbf{F}(t)$ is a $K \times 1$ vector of returns to factors from time $t-1$ to time t , and $\mathbf{u}(t)$ is an N vector of mean-zero stock specific returns from $t-1$ to t .

In a cross-section regression model the objective is to use a set of stock factor exposures to explain the cross-section dispersion of stock returns. At a given time t the factor returns can be estimated from the cross-sectional regression model in equation (1).

The stock exposures to a given factor are traditionally obtained by normalising the factor in the cross-section using a z-score transformation at each time $t-1$. In this way we get a collection of factor z-scores for each stock at each time $t-1$. These factor z-scores are used to construct the factor exposure matrix $\mathbf{B}(t-1)$.

At a given point in time the factor returns $\mathbf{F}(t)$ can then be calculated from the OLS equation:

$$\mathbf{F}(t) = [\mathbf{B}^T(t-1)\mathbf{B}(t-1)]^{-1}\mathbf{B}^T(t-1)\mathbf{R}(t) \quad (2)$$

This can be re-written as:

$$\mathbf{F}(t) = \mathbf{W}(t-1)\mathbf{R}(t) \quad (3)$$

Where $\mathbf{W}(t-1) = [\mathbf{B}^T(t-1)\mathbf{B}(t-1)]^{-1}\mathbf{B}^T(t-1)$ is the matrix with the stock weights required to form orthogonal factor portfolios with a unit exposure to a given factor and zero exposure to all other factors. Repeating the regression at all times t yields the time series of factor returns.

Forecasting stock excess returns with cross-sectional regression models

In practical uses of the model, equation (1) is transformed into a problem of forecasting stock excess returns. If we have a forecast for the factor returns today, $E[\mathbf{F}(t+1)]$, we can use this along with today's factor exposures $\mathbf{B}(t)$ in order to find the expected stock excess returns, $E[\mathbf{R}(t+1)]$:

$$E[\mathbf{R}(t+1)] = \mathbf{B}(t)E[\mathbf{F}(t+1)] \quad (4)$$

The forecast of the factor returns $E[\mathbf{F}(t + 1)]$ is typically derived from a trailing average of the factor returns, e.g. over the trailing 12 months:

$$E[\mathbf{F}(t + 1)] = \frac{1}{T} \sum_{i=0}^{T-1} \mathbf{F}(t - i) \quad (5)$$

This is the approach that has been employed by many quantitative managers (here somewhat simplified) when managing multi-factor portfolios. We used to manage multi-factor portfolios in this way until 2013. Note that equation (5) is essentially a momentum model on factor returns. The *weight* of factors in forecasting stock excess returns is thus dynamic as defined by the trailing average of factor returns. This approach performed poorly in 2008 and 2009 for a standard factor set including value, quality and momentum stock exposures, with the historical average of factor returns failing to forecast future factor returns. This is one of the reasons why we adopted a constant risk budgeting allocation to factors in our most recent multi-factor approaches, backed by our research as in the papers: R. Perchet, R. Leote de Carvalho, P. Moulin. “Inter-temporal risk parity: a constant volatility framework for factor investing.”, *Journal of Investment Strategies*, Vol. 4, No. 1 (2014), pp. 1-23, and: R. Leote de Carvalho, X. Lu, P. Moulin. “An integrated risk-budgeting approach for multi-strategy equity portfolios.” *Journal of Asset Management*, Vol. 15, No. 1 (February 2014), pp. 24-47.

Portfolio construction

The expected stock excess returns can be used in a simplified way with the final portfolio just an equally weighted portfolio constructed by keeping a number of stocks with the highest expected excess returns derived from (4) and (5). However, some managers use the expected stock excess returns as input in a mean-variance optimisation in order to build the final multi-factor portfolio. This is also the approach we used to follow until 2013 in our multi-factor approaches. But this final mean-variance optimisation step requires a set of portfolio constraints which must be included in order to avoid corner solutions. This is one of the reasons why we adopted the new framework based on estimating robust stock excess returns from factor exposures as described in our research paper: R. Leote de Carvalho, X. Lu, P. Moulin. “An integrated risk-budgeting approach for multi-strategy equity portfolios.” *Journal of Asset Management*, Vol. 15, No. 1 (February 2014), pp. 24-47. This framework avoids the need to impose portfolio constraints other than those imposed by guidelines thanks to the use of robust forecast of stock excess returns as inputs into the optimiser.

Cross-sectional regression models in practice

a. Univariate approach

In the univariate case the problem is simple. In this case, equation (2) simplifies to:

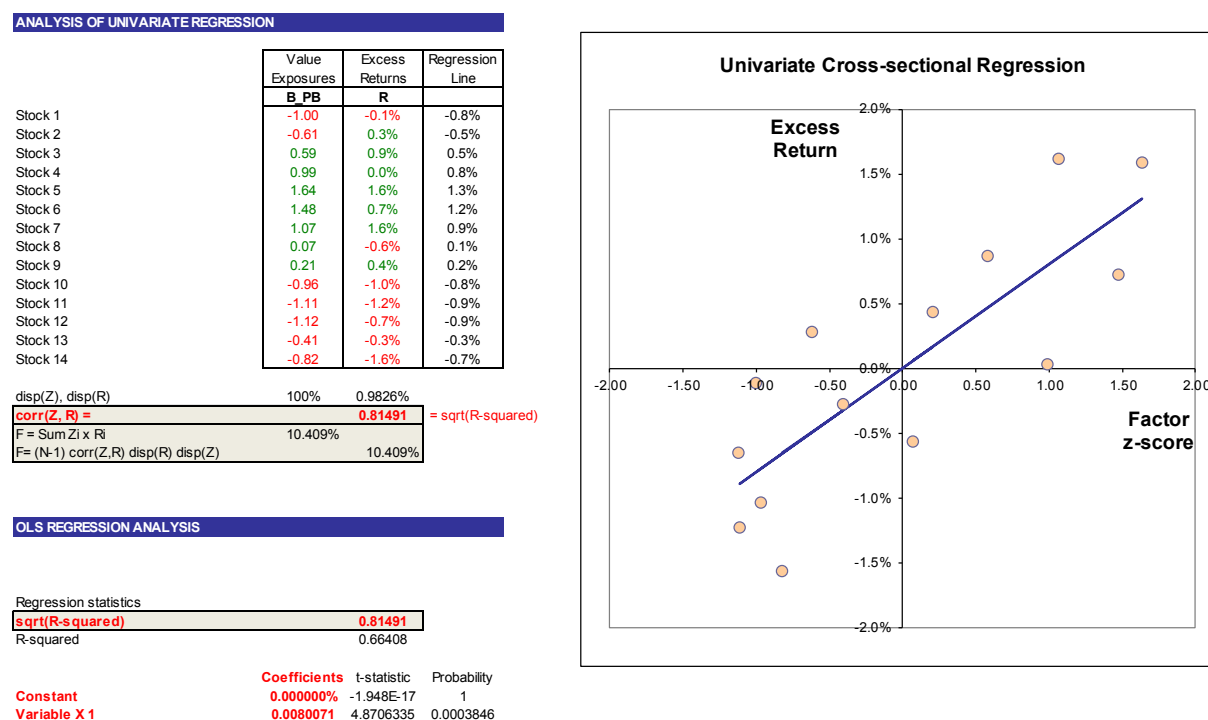
$$F = (N - 1) \text{corr}(\mathbf{Z}, \mathbf{R}) \text{disp}(\mathbf{Z}) \text{disp}(\mathbf{R}) \quad (6)$$

Where $corr(\mathbf{Z}, \mathbf{R})$ is the correlation between the factor z-scores $\mathbf{Z}$ and the stock excess returns $\mathbf{R}$ at that point in time, and is often known as the information coefficient. It is also the square root of the r-squared of the regression of the z-scores $\mathbf{Z}$ against the stock excess returns $\mathbf{R}$. $disp(\mathbf{Z})$ and $disp(\mathbf{R})$ are the cross-sectional dispersion, i.e. standard deviations, of $\mathbf{Z}$ and $\mathbf{R}$, respectively. In this case the factor returns can also be calculated from:

$$F = \sum_i Z_i R_i = (N - 1) corr(\mathbf{Z}, \mathbf{R}) disp(\mathbf{Z}) disp(\mathbf{R}) \quad (7)$$

In exhibit 1 we show the example of hypothetical z-scores obtained from the price-to-book of stocks at a given point in time. We call these *value exposures*. In the example $N = 14$ stocks in the universe. In the column in the middle we include hypothetical stock excess returns.

Exhibit 1: example of univariate cross-sectional regression at a given point in time.



a. Multi-variate approach with low correlated factors

In exhibit 2 we give the example of a three factor model. The hypothetical factors considered are price-to-book, size (stock market capitalisation) and momentum (stock historical returns). In the first three columns we include hypothetical factor z-scores for each stock. In the next three set of columns we show $\mathbf{W}(t - 1)$. The sum-product of these factor weights by the factor exposures gives the results on the bottom of the table. We can see that the first portfolio has unit exposure to price-to-book and zero exposure to the other two factors. And similarly for the other two factor portfolios. We compare the factor weights from the cross-sectional regression with simplified factor weights obtained from keeping the stocks with the largest factors z-scores against the stocks with the smallest factor

exposures, which can be found in the last three columns of the table. In the last column we include the excess returns of each stock used to estimate the factor returns as in (3). These are shown in the last row of the table, *Factor Returns*. The stock forecast returns can be found in the last column, *Forecast Returns*. They are found from the sum-product of the stock factor exposure by the factor returns as in (4), assuming here one period only in (5), i.e. that $T = 1$. In practical terms more than one period needs to be considered in (5) because otherwise not only too much turnover is generated, but also the results are too noisy with only one period. We used to consider 12 months as mentioned before.

In this example we can see that the orthogonal factor portfolio weights defined by $W(t - 1)$ are highly correlated with the stock factor exposures. This can be seen by comparing the first three columns with the factor exposures against the fourth to sixth column with the orthogonal factor weights.

Exhibit 2: example of implementation of three factor model at a given point in time using low correlated factors

Cross-sectional orthogonalisation of factor exposures														
	Factor Exposures Betas=zscores			Orthogonal Factor Portfolios			Equal Weighted Portfolios			Rankings			Excess Returns	Forecast Returns
	Value Exposures	Size Exposures	Momentum Exposures	Value Portfolio	Size Portfolio	Momentum Portfolio	Value Portfolio	Size Portfolio	Momentum Portfolio	Value Rank	Size Rank	Momentum Rank		
	B_PB	B_SIZE	B_RET	W_PB	W_SIZE	W_RET	W_PB	W_SIZE	W_RET	W_PB	W_SIZE	W_RET		
Stock 1	-1.00	-0.28	-0.06	-7.67%	-0.13%	-0.86%	-25%	0%	0%	3	6	9	-0.12%	-0.79%
Stock 2	-0.61	-0.28	-0.68	-4.12%	-2.93%	-6.37%	0%	0%	-25%	6	9	4	0.27%	-0.51%
Stock 3	0.59	-0.28	-0.29	5.85%	-5.02%	-3.61%	0%	0%	0%	10	7	7	0.86%	0.37%
Stock 4	0.99	-0.28	1.10	8.61%	-2.13%	8.14%	25%	0%	25%	11	5	13	0.02%	0.66%
Stock 5	1.64	-0.27	0.31	14.43%	-6.03%	1.05%	25%	25%	0%	14	11	10	1.58%	1.14%
Stock 6	1.48	-0.28	-0.97	13.75%	-9.42%	-9.86%	25%	-25%	-25%	13	3	3	0.71%	1.02%
Stock 7	1.07	3.47	-1.08	0.27%	26.70%	0.22%	25%	25%	-25%	12	14	1	1.61%	1.58%
Stock 8	0.07	-0.29	-0.40	1.55%	-4.01%	-4.28%	0%	-25%	0%	8	2	6	-0.58%	-0.01%
Stock 9	0.21	-0.29	2.48	1.30%	3.89%	20.38%	0%	-25%	25%	9	1	14	0.43%	0.08%
Stock 10	-0.96	-0.28	0.79	-7.78%	2.16%	6.40%	-25%	-25%	25%	4	4	12	-1.04%	-0.77%
Stock 11	-1.11	-0.26	0.73	-9.07%	2.62%	6.09%	-25%	25%	25%	2	12	11	-1.24%	-0.87%
Stock 12	-1.12	-0.28	-0.25	-8.61%	-0.32%	-2.38%	-25%	0%	0%	1	8	8	-0.66%	-0.88%
Stock 13	-0.41	-0.27	-1.07	-2.20%	-4.53%	-9.73%	0%	0%	-25%	7	10	2	-0.29%	-0.36%
Stock 14	-0.82	-0.14	-0.61	-6.29%	-0.85%	-5.20%	0%	25%	0%	5	13	5	-1.57%	-0.63%
Total				0.00%	0.00%	0.00%	0.00%	0.00%	0.00%					
Sum Longs				45.74%	35.37%	42.30%								
Sum Shorts				-45.74%	-35.37%	-42.30%								
Betas														
PB exposure				1.00	0.00	0.00	2.34	0.00	-0.60					
SIZE exposure				0.00	1.00	0.00	0.93	0.99	-0.94					
RET exposure				0.00	0.00	1.00	-0.46	-0.64	2.22					
				OLS			Ranking							
Factor Returns				0.733%	0.229%	-0.001%	1.745%	0.213%	-1.034%					

Excess Returns	Forecast Returns
R	R t
-0.12%	-0.79%
0.27%	-0.51%
0.86%	0.37%
0.02%	0.66%
1.58%	1.14%
0.71%	1.02%
1.61%	1.58%
-0.58%	-0.01%
0.43%	0.08%
-1.04%	-0.77%
-1.24%	-0.87%
-0.66%	-0.88%
-0.29%	-0.36%
-1.57%	-0.63%

b. Multi-variate approach with correlated factors

In exhibit 3 we repeat the example. However, in this case stocks have comparable exposures to the first two factors, price-to-book and price-to earnings, i.e. two *value factors*. As shown, the orthogonal factor weights for the two correlated value factors in columns four and five differ rather significantly from the original factor exposures in the first two columns. But this is the set of weights that yields unit exposures to each factor while neutralising the exposures to the other factors.

Clearly this poses a problem if we want to forecast stock excess returns using (4) and (5) using more than one period in (5). Since the factor weights are now very sensitive to correlations, applying (5) becomes tricky. First, it is difficult to know how a given set of orthogonalised factors at a given time is

related to the set of orthogonalised factors found in the previous period, invalidating the idea of using trailing averages of factor returns. Second, the factors themselves lose their meaning: what is exactly a factor with unit exposure to price-to-book but fully orthogonal to the highly correlated price-to-earnings indicator other than a complex mathematical construction over-relying on correlations?

Exhibit 3: example of implementation of three factor model at a given point in time using highly correlated factors

Cross-sectional orthogonalisation of factor exposures													
	Factor Exposures			Orthogonal Factor Portfolios			Equal Weighted Portfolios			Rankings			
	Value	Value	Betas=zscores	Value	Value	Momentum	Value	Value	Momentum	Value	Value	Momentum	
	Exposures	Exposures	Exposures	Portfolio	Portfolio	Portfolio	Portfolio	Portfolio	Portfolio	Rank	Rank	Rank	
	B_PB	B_PE	B_RET	W_PB	W_PE	W_RET	W_PB	W_PE	W_RET	W_PB	W_PE	W_RET	
Stock 1	-1.00	-1.08	-0.06	15.37%	-26.45%	-1.57%	-25%	-25%	0%	3	2	9	
Stock 2	-0.61	-0.45	-0.68	-12.68%	8.86%	-5.23%	0%	0%	-25%	6	6	4	
Stock 3	0.59	0.35	-0.29	22.76%	-21.03%	-2.67%	0%	0%	0%	10	10	7	
Stock 4	0.99	1.10	1.10	-21.24%	33.52%	9.75%	25%	25%	25%	11	12	13	
Stock 5	1.64	1.32	0.31	23.43%	-12.28%	2.55%	25%	25%	0%	14	14	10	
Stock 6	1.48	0.98	-0.97	48.06%	-42.41%	-8.19%	25%	25%	-25%	13	11	3	
Stock 7	1.07	1.11	-1.08	-8.14%	18.35%	-7.42%	25%	25%	-25%	12	13	1	
Stock 8	0.07	0.20	-0.40	-13.93%	16.44%	-2.58%	0%	0%	0%	8	9	6	
Stock 9	0.21	0.20	2.48	-7.75%	11.65%	19.53%	0%	0%	25%	9	8	14	
Stock 10	-0.96	-0.65	0.79	-30.08%	26.27%	6.48%	-25%	0%	25%	4	5	12	
Stock 11	-1.11	-1.55	0.73	52.96%	-70.26%	3.30%	-25%	-25%	25%	2	1	11	
Stock 12	-1.12	-0.84	-0.25	-22.34%	15.63%	-1.84%	-25%	-25%	0%	1	3	8	
Stock 13	-0.41	0.00	-1.07	-38.66%	40.32%	-7.20%	0%	0%	-25%	7	7	2	
Stock 14	-0.82	-0.69	-0.61	-7.74%	1.39%	-4.91%	0%	-25%	0%	5	4	5	
Total				0.00%	0.00%	0.00%	0.00%	0.00%	0.00%				
Sum Longs				162.57%	172.43%	41.60%							
Sum Shorts				-162.57%	-172.43%	-41.60%							
Betas													
PB exposure				1.00	0.00	0.00	2.34	2.31	-0.60				
PE exposure				0.00	1.00	0.00	2.16	2.17	-0.63				
RET exposure				0.00	0.00	1.00	-0.46	-0.11	2.22				
Cross-Sectional correlation (B_PB, B_PE):				0.96									
Cross-Sectional correlation (W_PB, W_PE):				-0.96									
				OLS			Ranking						
Factor Returns				0.806%	-0.009%	-0.071%	1.745%	1.878%	-1.034%				

Excess Returns	Forecast Returns	
R1-1	R1	
-0.12%	-0.79%	15%
0.27%	-0.44%	0%
0.86%	0.49%	23%
0.02%	0.71%	0%
1.58%	1.28%	23%
0.71%	1.25%	48%
1.61%	0.93%	0%
-0.58%	0.08%	0%
0.43%	-0.01%	0%
-1.04%	-0.83%	0%
-1.24%	-0.93%	53%
-0.66%	-0.88%	0%
-0.29%	-0.25%	0%
-1.57%	-0.61%	0%

for regression

Dampening the off-diagonal elements in the covariance matrix $B^T(t-1)B(t-1)$ is a common practical solution to the problem. In exhibit 4 we show the impact of multiplying the off-diagonal elements by 20%, strongly dampening their impact in the factor orthogonalisation. In this case we can see that the factor weights resembles again the factor exposures but we end up increasing the weight of the value factors in the final calculation of expected returns, which can be seen from the factor returns at the bottom of the table. Due to the dampening of the correlations, the expected stock excess returns have a stronger weight coming from value exposures than before dampening. This is how we used to solve this problem until 2013.

Exhibit 4: example of implementation of three factor model at a given point in time using highly correlated factors and dampening the covariance matrix

Cross-sectional orthogonalisation of factor exposures

	Factor Exposures Betas=z-scores			Orthogonal Factor Portfolios			Equal Weighted Portfolios			Rankings			Excess Returns	Forecast Returns
	Value Exposures	Value Exposures	Momentum Exposures	Value Portfolio	Value Portfolio	Momentum Portfolio	Value Portfolio	Value Portfolio	Momentum Portfolio	Value Rank	Value Rank	Momentum Rank		
	B_PB	B_PE	B_RET	W_PB	W_PE	W_RET	W_PB	W_PE	W_RET	W_PB	W_PE	W_RET		
Stock 1	-1.00	-1.08	-0.06	-6.14%	-8.77%	-0.66%	-25%	-25%	0%	3	2	9	-0.12%	-1.43%
Stock 2	-0.61	-0.45	-0.68	-4.18%	-3.35%	-5.35%	0%	0%	-25%	6	6	4	0.27%	-0.67%
Stock 3	0.59	0.35	-0.29	4.07%	2.38%	-2.19%	0%	0%	0%	10	10	7	0.86%	0.67%
Stock 4	0.99	1.10	1.10	6.11%	9.11%	8.63%	25%	25%	25%	11	12	13	0.02%	1.35%
Stock 5	1.64	1.32	0.31	10.85%	10.06%	2.58%	25%	25%	0%	14	14	10	1.58%	2.01%
Stock 6	1.48	0.98	-0.97	10.10%	6.86%	-7.28%	25%	25%	-25%	13	11	3	0.71%	1.78%
Stock 7	1.07	1.11	-1.08	6.60%	8.80%	-8.11%	25%	25%	-25%	12	13	1	1.61%	1.60%
Stock 8	0.07	0.20	-0.40	0.19%	1.80%	-3.04%	0%	0%	0%	8	9	6	-0.58%	0.23%
Stock 9	0.21	0.20	2.48	1.42%	1.89%	19.13%	0%	0%	25%	9	8	14	0.43%	0.06%
Stock 10	-0.96	-0.65	0.79	-6.55%	-4.60%	5.93%	-25%	0%	25%	4	5	12	-1.04%	-1.18%
Stock 11	-1.11	-1.55	0.73	-6.22%	-13.05%	5.42%	-25%	-25%	25%	2	1	11	-1.24%	-1.91%
Stock 12	-1.12	-0.84	-0.25	-7.52%	-6.31%	-2.06%	-25%	-25%	0%	1	3	8	-0.66%	-1.33%
Stock 13	-0.41	0.00	-1.07	-3.30%	0.54%	-8.22%	0%	0%	-25%	7	7	2	-0.29%	-0.18%
Stock 14	-0.82	-0.69	-0.61	-5.43%	-5.35%	-4.78%	0%	-25%	0%	5	4	5	-1.57%	-0.99%
Total				0.00%	0.00%	0.00%	0.00%	0.00%	0.00%					
Sum Longs				39.35%	41.44%	41.69%								
Sum Shorts				-39.35%	-41.44%	-41.69%								
Betas														
PB exposure				0.85	0.88	-0.02	2.34	2.31	-0.60					
PE exposure				0.73	0.85	-0.05	2.16	2.17	-0.63					
RET exposure				-0.02	-0.06	1.00	-0.46	-0.11	2.22					
Cross-Sectional correlation (B_PB, B_PE):				0.96										
Cross-Sectional correlation (W_PB, W_PE):				0.92										
				OLS			Ranking							
Factor Returns				0.677%	0.705%	-0.089%	1.745%	1.878%	-1.034%					

Conclusions:

This three factor example illustrates the difficulties of implementing cross-sectional regression models in practice. The orthogonalisation of factors can lead to some rather unintuitive orthogonalised unit exposure factor portfolios in the presence of correlated factors. Dampening correlations reduces the problem but increases the weight of correlated factors in the final calculation of expected stock excess returns. If we damp the correlations completely then the final stock expected excess returns become a simple weighted average of stock factor z-scores, weighted by the factor expected returns, typically calculated from trailing averages of factor returns. The dampening of correlations defeats the purpose of using cross-sectional regressions.

The question of relying on trailing averages of factor returns in order to define factor weights can also be questioned. That will typically lead to negative exposure to factors after they performed poorly. Our research on factor volatility and negative correlation between factor volatility and factor premiums establish a stronger theoretical case for defining the risk budgeting allocation to factors.

Using the stock expected returns derived in this way in a final mean-variance optimisation step is also not efficient in our view as the optimal portfolios typically need to be strongly constrained. The use of robust expected stock excess returns offers a solution to this problem: only guidelines constraints are considered and no additional constraints are required since corner solutions are avoided with this approach.

After many years using a framework along the lines here described, based on cross-sectional regressions, with strong dampening of correlations and using a mean-variance optimiser to build the final portfolio, we have finally changed the investment process in 2013 in order to take into account our most recent research in terms of combining factors efficiently and properly taking into account

their correlations while allocating a well-defined risk budgeting to each factor, while avoiding relying on a simplified factor return momentum model, and replacing the optimisation step by an improved optimisation step based on robust forecasts of stock excess returns derived from factor exposures.